

Detecting photo reuse under generative image edits

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Research question and DoorDash opportunity

Can an image-reuse detector remain reliable after diffusion-based edits while distinguishing genuinely independent photographs of visually similar food? I propose a benchmark and retrieval-verification system that separates shared photographic origin from shared semantic content, with explicit control of false matches and review cost.

A DoorDash 2024 intern project used vision-transformer embeddings to detect reused refund photos, extending beyond exact and perceptual hashes to handle transformations such as rotation and scaling.^[1] I would investigate a further robustness question: how does that class of detector behave when generative editing changes a background, removes an object, or relights part of a photograph? This is a proposed threat model, not a claim about the prevalence of these edits in DoorDash traffic.

My hypothesis is that a fast global embedding retriever followed by local, source-consistency verification will improve recall at a fixed false-match rate. The difficult case is not simply detecting AI-generated content. Two independent photos of the same dish should remain distinct, while an edited derivative may retain enough source-specific evidence to be recognized as reuse. Uncertain pairs should be routed to review.

Approach and research contribution

Build a lineage-controlled benchmark. With approved data access, assemble genuine source photos, independent same-dish photographs, and known derivatives. Produce controlled edits spanning conventional transformations and several generative edit types and strengths. Store the source identifier, edit mask, model revision, prompt, and seed. Distinguish known derivation from visually verifiable reuse: source lineage alone does not make a heavily regenerated image an identifiable copy.

Compare meaningful baselines. Evaluate exact hashes, perceptual hashes, a generic vision-transformer embedding, and a dedicated copy-detection descriptor such as SSCD.^[2] Include a diffusion-copy baseline where reproducible code permits. ICDiff already studies diffusion-generated replication; this proposal extends that question to food imagery with difficult same-item negatives, operational review budgets, and an explicitly held-out editor.^[3]

Retrieve broadly and verify selectively. Use embeddings to retrieve candidate source photos, then check local patch correspondences and spatial consistency on the most plausible matches. A lightweight calibrated verifier would combine global similarity, locally supported overlap, and disagreement signals. Limit verification to a small candidate set and produce an evidence view showing matched regions. Reuse is an input to investigation, not proof of fraud or a basis for automatic refund denial.

The core research artifact would be a benchmark that distinguishes semantic resemblance from photographic reuse, paired with a reproducible study of when local verification helps. If a pretrained descriptor already meets the target, the useful result is a simpler operating policy and a documented failure map, rather than an unnecessarily complex model.

Evaluation and success criteria

Split by source photo before generating derivatives, and keep merchants or capture sessions disjoint where possible. Deduplicate sources across splits. Hold out at least one editor family and edit type to test generalization beyond the synthetic construction process. Keep uncertain or near-total regenerations in a separately adjudicated category. If available, reserve real reviewed cases as an external test; synthetic robustness alone cannot establish operational effectiveness.

Measure retrieval recall at a fixed candidate budget and end-to-end recall at a preregistered false-match ceiling. Estimate that ceiling on a large pool of independent same-dish and same-packaging negatives, with confidence intervals; choose sample size to support the intended claim. Report precision under realistic reuse prevalence, review coverage, abstention rate, and performance by edit strength and input quality. Do not infer business savings from a balanced synthetic test set.

Replay candidate searches against a fixed-size reference gallery. Report p50/p95 latency, embedding and index memory, verification cost, and reviewer workload. Ablate local verification, diffusion-derived calibration examples, and the abstention mechanism. Compare any learned verifier against a thresholded pretrained baseline under the same compute and labeling budgets. Use blinded adjudication for ambiguous pairs and preserve reviewer disagreement.

Success means recovering more identifiable reused images at the same false-match and review budgets, including on the held-out editor. A negative result would identify where generative edits destroy enough source evidence that the system should abstain. The deliverable would support better review decisions without claiming that all reused images are fraudulent or that all generated images can be detected.

Preparation and why DoorDash

At Purdue's EcoAI Lab, advised by Prof. Haoran You, my master's research focuses on efficient diffusion execution under memory and quality constraints. I have built configuration-driven benchmark runners, analyzed PyTorch and CUDA profiles on A100, and enabled quantized FLUX.2 [klein] generation on Jetson Orin Nano. That experience prepares me to build a controlled, reproducible editing pipeline and measure the computational cost of the proposed detector. Retrieval and verification would be a deliberate extension of my current systems work.

DoorDash offers a setting where similar-looking objects are common and mistaken matches have real consequences. Governed access to distinct captures, reviewed cases, and operational review constraints would make this study more informative than a generic copy benchmark. The earlier intern project provides a concrete technical starting point.^[1] I would seek a sponsor in Marketplace Integrity and collaborate with vision researchers on annotation and evaluation, aligning the work with the fellowship's emphasis on measurement in real operational settings.

Twelve-week plan and outcomes

Weeks 1-2: agree on scope, data access, labels, and false-match targets. **Weeks 3-5:** construct the benchmark and reproduce pretrained baselines. **Weeks 6-8:** implement local verification and calibrate abstention. **Weeks 9-12:** run held-out-editor and real-case evaluation, stress-test retrieval cost, and deliver a benchmark specification, reviewer-facing prototype, and research report. If internal labels arrive late, start with licensed public images and clearly limit claims until operational validation is possible.

Selected references

- [1] DoorDash. Part 1 of the 2024 summer intern projects. Photo recycling detection.
- [2] Pizzi et al. A Self-Supervised Descriptor for Image Copy Detection. CVPR 2022.
- [3] Wang et al. Image Copy Detection for Diffusion Models. NeurIPS 2024.